Venkata Lakshmi Parimala Pasupuleti

Agentic AI Engineer · LangGraph Multi-Agent · HITL & Confidence Scoring · RAG Pipelines · LLM Evaluation · Production Python

venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | AgentProbe (live) | Virginia, USA

PROFESSIONAL SUMMARY

Agentic AI engineer who builds systems where AI doesn't just respond — it reasons, executes, and knows when to stop. Built **Leasel**: an 8-agent LangGraph StateGraph with confidence-gated HITL escalation, deterministic compliance gates, and a core design rule enforced in code: *LLM reasons, deterministic logic or a human makes every regulated call*. Built **AgentProbe**: a CI-integrated 5-metric evaluation platform that scores every agent run on factual accuracy, hallucination, retrieval, completeness, and compliance — regressions blocked before merge. Deployed agentic workflows for real client verticals (marketing, real estate, commercial survey) at Magizhs. Shipped Entrata's first production AI pipeline across 1.6M financial records with compliance gates from day one. 8+ years production Python. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Agentic AI Systems | LangGraph Multi-Agent | LLM Orchestration | HITL & Confidence Scoring | RAG Pipelines | LLM Evaluation Frameworks | Multi-Step Workflow Execution | Tool Calling & Function Calling | Production Python | Automate vs. Escalate Design | Customer-Facing AI Deployment | CI-Integrated Agent Evals

EXPERIENCE

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer — Customer-Facing Agentic AI · LLM Orchestration · Multi-Step Workflow Execution

- Prototyped and shipped a **production agentic system directly with client verticals** (marketing, real estate, commercial survey) — orchestrated as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing), translating unstructured natural-language inputs into structured executable workflows.
- Built **multi-step execution logic** from requirements to production: photogrammetry math (ground sampling distance, camera footprint, overlap-driven line spacing), RAG over FAA Part 107 corpus, and deterministic guardrails enforced before any downstream action fires.
- Implemented **confidence-aware routing**: compliant workflows route to export; non-compliant ones route to rejection with grounded regulatory citations — the agent knows when to act and when to stop.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

ML Research · Statistical Modeling · Anomaly Detection

- Built prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection across 3 research pipelines surfacing data integrity issues manual review missed.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer — Production AI · Automation Workflows · Data Pipelines at Scale

- **Shipped Entrata's first production AI pipeline** across 1.6M real financial records (Python, PyTorch, OpenAI API) — automated complex enterprise workflows replacing manual Excel operations; output validation, quality gates, and compliance controls enforced at launch, zero violations.
- Built and shipped **20+ full-stack production modules** (Python, PHP, PostgreSQL, RabbitMQ) for 100K+ monthly active users across 120+ Agile sprints — 99.5% uptime sustained; zero security incidents across all deployments.
- **Eliminated 80% of manual operational workload** by designing and automating Python/Bash workflow pipelines; optimized 200+ SQL queries (30% performance gain); mentored 5+ engineers across 300+ peer code reviews.
- Achieved **100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines; led microservices architecture redesign (Docker, 50% scalability gain).

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Backend · Data Pipelines · AWS S3

- Built full-stack backend modules (PHP, PostgreSQL, JavaScript); processed 1.6M HUD/Tax Credit records; migrated client documents to AWS S3 and redesigned legacy database architecture.

PORTFOLIO PROJECTS

Leasel — 8-Agent LangGraph Platform with Confidence-Gated HITL & Evaluation Agentic AI HITL Confidence Scoring

2026 · Python, LangGraph (8-agent), FastAPI, TF-IDF, OCR, SQLite · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Designed an **8-agent LangGraph StateGraph** (Triage → Retrieval → Config → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) with real conditional edges — the system determines when to automate vs. escalate based on confidence scoring at every stage.
- Core **automate-vs-escalate design rule enforced in code**: LLM extracts and reasons; when confidence is below threshold, deterministic logic or human review takes over — never the reverse. 0 safety boundary violations on full eval set.
- Built end-to-end **RAG pipeline**: document ingestion (OCR-fallback) → section-aware chunking → TF-IDF retrieval → LLM reasoning → citation-grounded output. 100% citation correctness, 0% unsupported-claim rate on Phase 1 eval; correctly traces false-ineligibility ticket to misconfigured setting end-to-end.

AgentProbe — CI-Integrated LLM Evaluation & Multi-Agent Dev Platform LLM Eval Multi-Agent Learning Workflows

2024–Present · Python, LangGraph, LangChain, FastAPI, ChromaDB, Anthropic, OpenAI, Groq, Ollama, React, Docker, GitHub Actions · github.com/venkatalakshmiparimala/AgentProbe

- Built a **CI-integrated 5-metric evaluation framework** (factual accuracy, hallucination, retrieval, completeness, compliance) — every agent commit scored automatically via GitHub Actions; regressions blocked before merge; confidence signals tracked per-run.
- Deployed a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with tool-use patterns, inter-agent state handoffs, and **multi-provider LLM routing** (Anthropic, OpenAI, Groq, Ollama) — 15-endpoint FastAPI backend, trace-level observability via React dashboard.
- ChromaDB RAG knowledge pipeline backing retrieval-quality evaluation across concurrent agent test suites — per-run metrics, drift detection, latency, and retrieval quality tracked continuously.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Agentic AI & Orchestration: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, LLM Orchestration, HITL, Human-in-the-Loop, Confidence Scoring, Automate vs. Escalate Design, Tool Calling, Agent Workflow Integration, Learning Workflows, Context Engineering, Ethical AI, Safety Guardrails

LLM Evaluation & RAG: LLM Evaluation, LLM-as-Judge, Eval Frameworks, RAG Pipelines, Semantic Search, Embeddings, ChromaDB, Vector Database, Multi-Provider LLM Routing, OpenAI API, Anthropic API, Groq, Ollama, Prompt Engineering, GenAI, Foundation Model APIs

Backend & Infrastructure: Python (primary), FastAPI, REST APIs, AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), PostgreSQL, Microservices, Data Pipelines, ETL, RabbitMQ, Linux, Git

ML & Data: PyTorch, Scikit-learn, Isolation Forest, Random Forest, NLP, spaCy, BERT, NumPy, Pandas, Statistical Modeling, Feature Engineering, Anomaly Detection